

The Eyes of Texas are Upon OB/GYNs: Physician Migration and Crowdsourced Enforcement of Abortion Regulations

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Motivation: Criminal vs. Civil Liability

	Criminal	Civil
Damages who?	Society	Individual

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Damages who?	Society	Individual
Burden of Proof?	Beyond a reasonable doubt	A preponderance of evidence (50%)
Rights?	Innocent until proven guilty Right to an attorney No self-incrimination Witnesses in your defense Confront accusing witnesses	None

Motivation: What is a Civil Liability Enforcement Mechanism (CLEM)?

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Motivation: What is a Civil Liability Enforcement Mechanism (CLEM)?

- Attempts by a state to circumvent federal law will be ruled unconstitutional
- What if there's no gov't official to sue for unconstitutionality?
- Private citizens enforce through civil lawsuits; “bounty laws”
- “Not only unusual, but unprecedented... it is the role of the Supreme Court in our constitutional system that is at stake.” (SCOTUS Chief Justice John Roberts)
- Arms race on sensitive issues: **abortion in TX**, guns in CA, more: [CLEM Copycats](#)

Today's Case Study: Texan OB/GYNs and Abortion Law

Research Question: What is the effect of civil liability in the context of migration responses to abortion regulations?

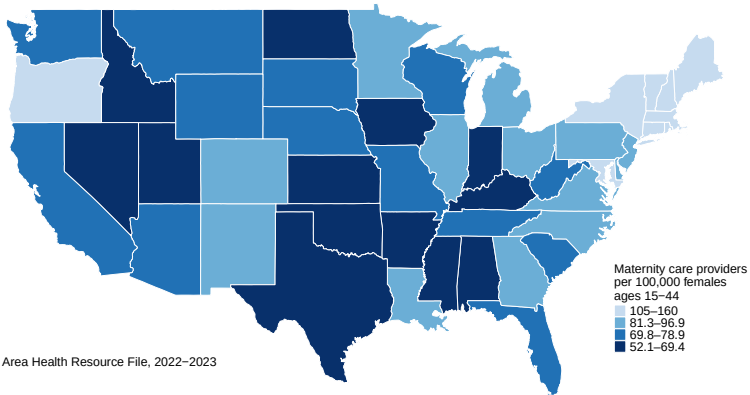
- Lower burden of proof and broader set of affected physicians
 - Criminal targets abortion performers; civil targets anyone who “aids or abets”
 - Omnipresent threat of scrutiny, hearsay
- Changes the physician optimization problem: increases expected operating costs
 - Malpractice insurance does not cover civil (or criminal) liability

Approach

- Triple difference model (also SC, S-DiD)
 - Texas - all other states
 - OB/GYNs - all other MDs/DOs
 - After SB 8 - Before SB 8
- Panel of Medicare administrative data from November 2007 (start of data) through June 2022 (*Dobbs* overturns *Roe v. Wade*)

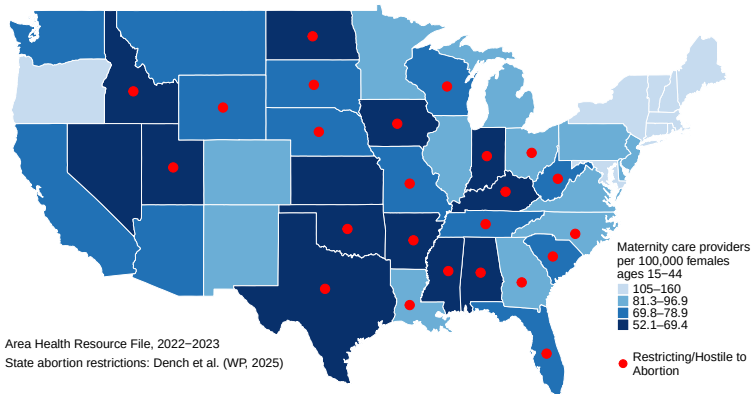
Preview of Results

States with criminal abortion restrictions tend to have fewer OB/GYNs



Area Health Resource File, 2022–2023

States with criminal abortion restrictions tend to have fewer OB/GYNs



Maternal Mortality

Criminal penalties decrease OB/GYN supply

- TRAP (Targeted Regulation of Abortion Providers) era, 1993-2022: [Timeline of TX Abortion Law](#)
 - 5.3% decrease over 28 years in OB/GYNs in TRAP states [Markowski & Vandenbroeck](#)
(*Working Paper*, 2025)

Criminal penalties decrease OB/GYN supply

- TRAP (Targeted Regulation of Abortion Providers) era, 1993-2022: [Timeline of TX Abortion Law](#)
 - 5.3% decrease over 28 years in OB/GYNs in TRAP states [Markowski & Vandenbroeck \(Working Paper, 2025\)](#)
- Post-*Dobbs* era, 2022-present:
 - 3% decrease over 3 years in OB/GYNs in ban states [Diaz-Campo & Pineda-Torres \(Working Paper, 2025\)](#)
 - 0.049% decrease in total population in ban states (52,600/quarter) [Dench et al. \(Working Paper, 2025\)](#)

Contribution: But how does civil liability enforcement compare to criminal penalties?

Texas Senate Bill 8: “Heartbeat Bill”

- Took effect in 2021, **during** *Roe v. Wade* era: [Timeline of TX Abortion Law](#)
- Bans abortion after detection of fetal heartbeat (6 weeks)
- Civil Liability Enforcement: Minimum \$10,000 in damages against any who “aid or abet” an abortion, no fee recovery for defendant

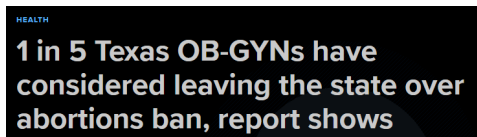
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- Bans abortion after detection of fetal heartbeat (6 weeks)
- Civil Liability Enforcement: Minimum \$10,000 in damages against any who “aid or abet” an abortion, no fee recovery for defendant
- Strategic inaction $\implies$ avoid creating test cases $\implies$ survives the court system

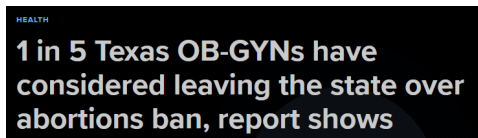
SB 8 is Salient

- Preserved deterrent effect, minimized risk of invalidating SB-8
- Three suits filed against Dr. Alan Braid, dismissed 2.5 yrs later

People are saying...



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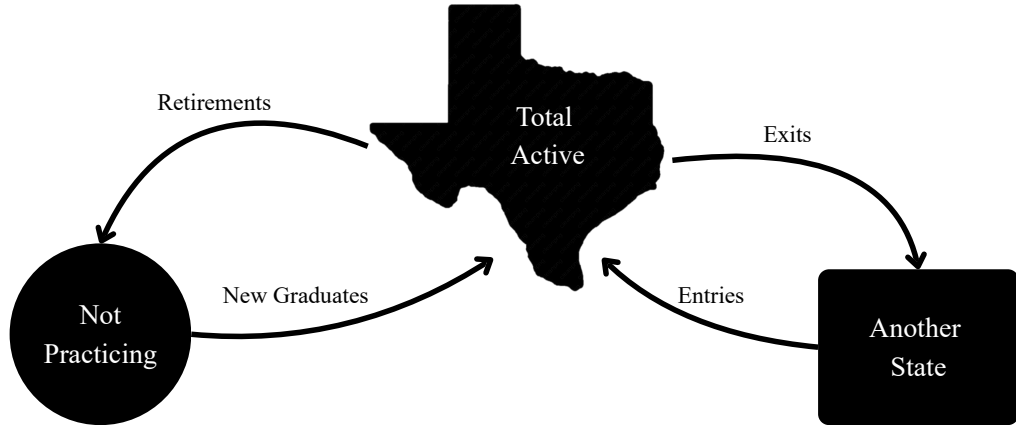
Cost to switch states: $\approx$ \$150k to \$250k

So what does the data show?

Data

- Novel quarterly panel of the universe of U.S. physicians (NPPES and DAC):
 - Physician name & NPI
 - Practice street address (must be updated within 30 days of moving)
 - Specialty
- Includes existing physicians and new physicians coming from residency
- Period: Nov. 2007 to June 2022, $N \approx 1.4\text{m}$ physicians
- Migration is defined as:
 - entry into TX, exit out of TX
 - retirement (exiting practice), new graduates (entering practice)

Outcome Variables - Migration



Summary (by state-specialty-quarter)

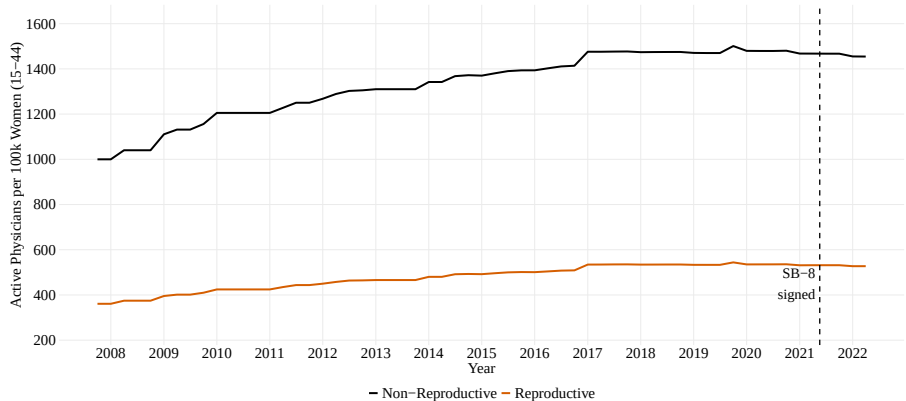
	Mean	SD	Min	Max
Active OB/GYN	92.86	22.96	44	216
Active all other physicians	53.41	100.33	0	1248
Enter state OB/GYN	1.36	2.24	0	44
Exit state OB/GYN	0.92	1.78	0	23
Enter state all physicians	0.87	3.28	0	263
Exit state all physicians	0.60	2.62	0	129

Trend: Reproductive Providers vs. Other Physicians, all states

OB/GYN

ER

Family

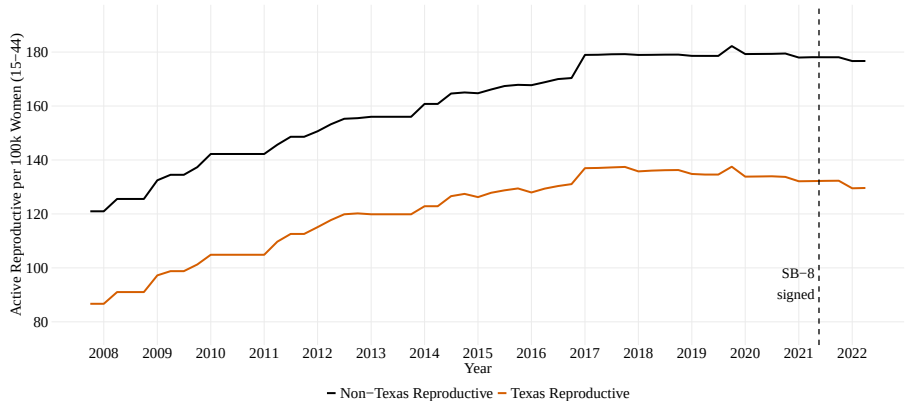


Trend: Reproductive Providers outside vs. inside Texas

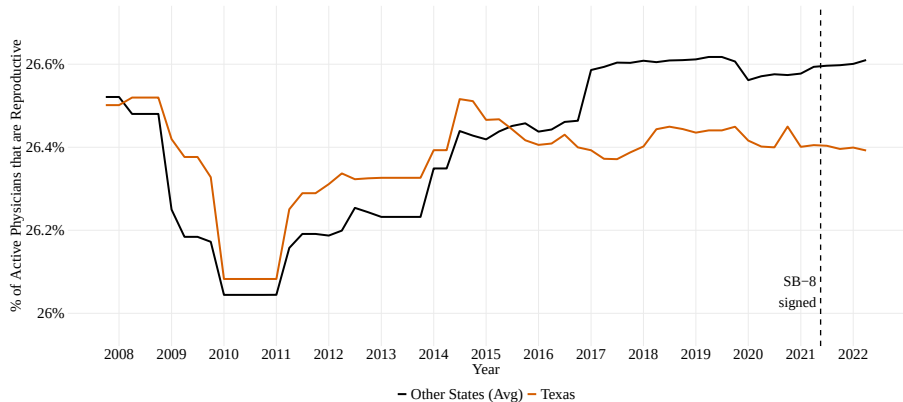
OB/GYN

ER

Family



Percentage of Active Physicians: Reproductive Providers



OB/GYN

ER

Family

SB-8
signed

Descriptive Means

	Not TX (Pre)	TX (Pre)	Not TX (Post)	TX (Post)
OB/GYN: Active	91.63	79.21	98.20	79.75
OB/GYN: Enter state	1.38	1.08	1.35	0.78
OB/GYN: Exit state	0.73	0.38	1.64	1.01
All other physicians: Active	52.95	37.32	56.35	38.71
All other physicians: Enter state	0.85	0.58	0.89	0.54
All other physicians: Exit state	0.45	0.20	1.09	0.71

Triple Diff: Time, State, & Specialty

Identifying assumption: Had Texas not passed SB 8, OB/GYNs in Texas would behave similarly to OB/GYNs in other states before and after May 19, 2021

$$\text{Outcome}_{ist} = \alpha + \beta_1 \cdot \underbrace{\text{Post SB8}_t}_{=1 \text{ if after Sept 1, 2021}} \times \underbrace{\text{Texas}_s}_{=1 \text{ if Texas}} \times \underbrace{\text{OB/GYN}_i}_{=1 \text{ if OB/GYN}} + \bar{\beta} \cdot (\text{Interaction Terms})$$

+ $\underbrace{u_{ist}}_{\text{error term}}$

where $\text{Outcome}_{ist} \in \{\text{Total Active, Exit, Entry, Enter Practice, Exit Practice}\}$

Triple Diff Results

	Active	Exit	Enter	Exit	Enter
		State	State	Practice	Practice
β_1	-6.33	0.23	0.25	0.01	0.08
	(26.36)	(0.45)	(0.79)	(0.12)	(0.58)
N	100,834	100,834	99,183	100,834	99,183

OLS. SE in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

[Only OB/GYN](#)[Only ER](#)[Only Family](#)

	Active	Exit State	Enter State	Exit Practice	Enter Practice
TX	-12.79*** (2.23)	-0.21*** (0.04)	-0.22** (0.07)	-0.01 (0.01)	-0.06 (0.05)
Reproductive	115.32*** (1.08)	0.76*** (0.02)	1.66*** (0.03)	0.07*** (0.005)	0.94*** (0.02)
Post	3.35** (1.10)	-0.16*** (0.02)	-0.51*** (0.03)	-0.02*** (0.005)	-0.35*** (0.02)
TX × Repro	-26.16*** (7.68)	-0.45*** (0.13)	-0.45 (0.23)	-0.01 (0.03)	-0.08 (0.17)
TX × Post	-1.07 (7.62)	0.08 (0.13)	0.13 (0.23)	0.01 (0.03)	0.06 (0.17)
Repro × Post	15.10*** (3.70)	-0.38*** (0.06)	-1.22*** (0.11)	-0.07*** (0.02)	-0.93*** (0.08)
β_1	-6.33 (26.36)	0.23 (0.45)	0.25 (0.79)	0.01 (0.12)	0.08 (0.58)
Constant	43.75*** (0.32)	0.39*** (0.01)	0.72*** (0.01)	0.02*** (0.001)	0.35*** (0.01)
N	100,834	100,834	99,183	100,834	99,183

SE in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Synthetic Diff-in-Diff: Reproductive Providers

	Active	Exit	Enter	Exit	Enter
		State	State	Practice	Practice
ATT	2.69	-0.53	-0.52	-0.0000	-0.002

Synth DiD (Arkhangelsky et al. 2021). SE from placebo.

Only OB/GYN

Only ER

Only Family

Discussion

Limitations:

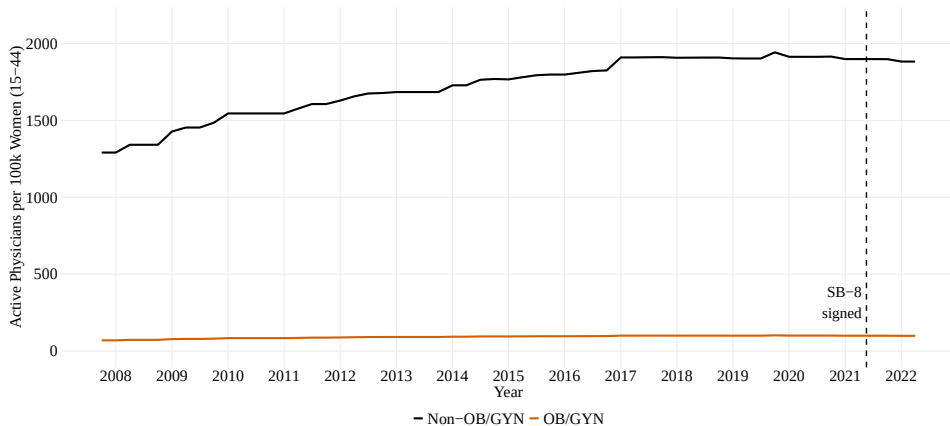
- Low statistical power: one treated state, five treated quarters

Next Steps:

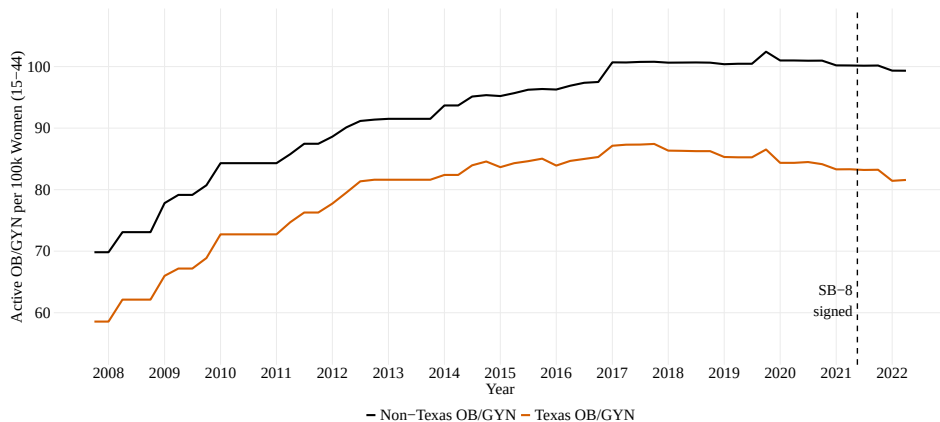
- Event study
- Heterogeneity analysis: urban vs. rural counties, physician age
- How to observe retirement from practice?

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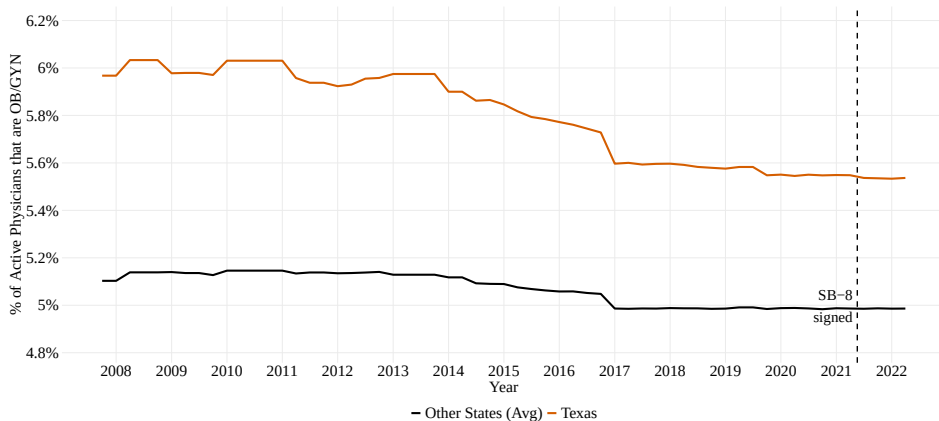
Trend: OB/GYNs vs. Other Physicians



Trend: OB/GYNs outside vs. inside Texas



Percentage of Active Physicians: OB/GYN



Descriptive Means: OB/GYN Only

	Not TX (Pre)	TX (Pre)	Not TX (Post)	TX (Post)
OB/GYN: Active	91.63	79.21	98.20	79.75
OB/GYN: Enter state	1.38	1.08	1.35	0.78
OB/GYN: Exit state	0.73	0.38	1.64	1.01
All other physicians: Active	52.95	37.32	56.35	38.71
All other physicians: Enter state	0.85	0.58	0.89	0.54
All other physicians: Exit state	0.45	0.20	1.09	0.71

Results: OB/GYN Only

	Active	Exit	Enter	Exit	Enter
		State	State	Practice	Practice
β_1	-3.43	0.06	-0.03	-0.01	-0.11
	(46.98)	(0.76)	(1.35)	(0.20)	(0.98)
N	100,834	100,834	99,183	100,834	99,183

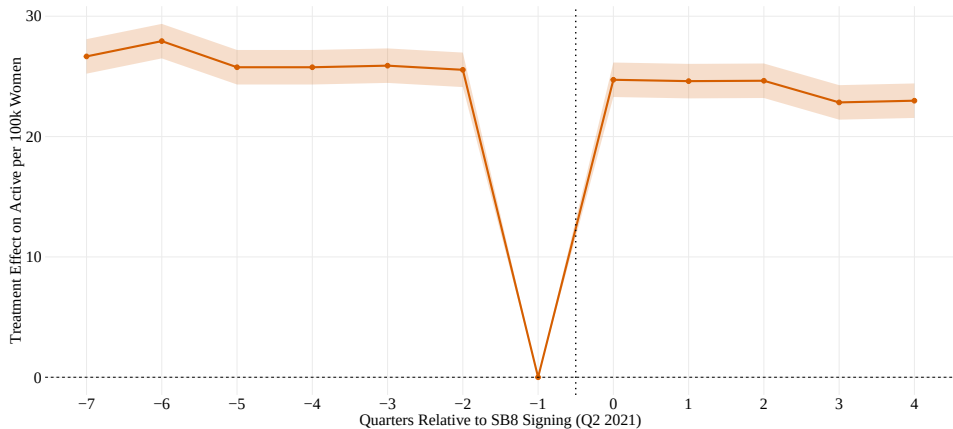
OLS. SE in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

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Full Results: OB/GYN Only

	Active	Exit	Enter	Exit	Enter
		State	State	Practice	Practice
TX	-15.63*** (2.30)	-0.25*** (0.04)	-0.27*** (0.07)	-0.01 (0.01)	-0.08 (0.05)
OB/GYN	38.68*** (1.92)	0.28*** (0.03)	0.53*** (0.06)	0.02* (0.01)	0.25*** (0.04)
Post	4.35*** (1.13)	-0.20*** (0.02)	-0.61*** (0.03)	-0.03*** (0.005)	-0.43*** (0.02)
TX $\times$ OB/GYN	3.21 (13.68)	-0.11 (0.22)	-0.04 (0.40)	0.01 (0.06)	0.11 (0.29)
TX $\times$ Post	-1.44 (7.84)	0.10 (0.13)	0.16 (0.22)	0.01 (0.03)	0.08 (0.16)
OB/GYN $\times$ Post	3.86 (6.58)	-0.11 (0.11)	-0.35 (0.19)	-0.02 (0.03)	-0.25 (0.14)
β_1	-3.43 (46.98)	0.06 (0.76)	-0.03 (1.35)	-0.01 (0.20)	-0.11 (0.98)
Constant	52.95***	0.45***	0.85***	0.03***	0.43***

Event Study: Active OB/GYN



Synthetic Diff-in-Diff: OB/GYN Only

	<i>Dependent variable:</i>				
	OB/GYN (Per 100k Women)				
	Active	Exit State	Enter State	Exit Practice	Enter Practice
	(1)	(2)	(3)	(4)	(5)
ATT	−0.44	−0.31	−0.19	−0.0000	0.0005

Descriptive Means: ER Only

	Not TX (Pre)	TX (Pre)	Not TX (Post)	TX (Post)
OB/GYN: Active	91.63	79.21	98.20	79.75
OB/GYN: Enter state	1.38	1.08	1.35	0.78
OB/GYN: Exit state	0.73	0.38	1.64	1.01
All other physicians: Active	52.95	37.32	56.35	38.71
All other physicians: Enter state	0.85	0.58	0.89	0.54
All other physicians: Exit state	0.45	0.20	1.09	0.71

Results: ER Only

	Active	Exit State	Enter State	Exit Practice	Enter Practice
β_1	-1.71 (46.92)	0.10 (0.76)	0.04 (1.35)	-0.001 (0.20)	0.02 (0.98)
N	100,834	100,834	99,183	100,834	99,183

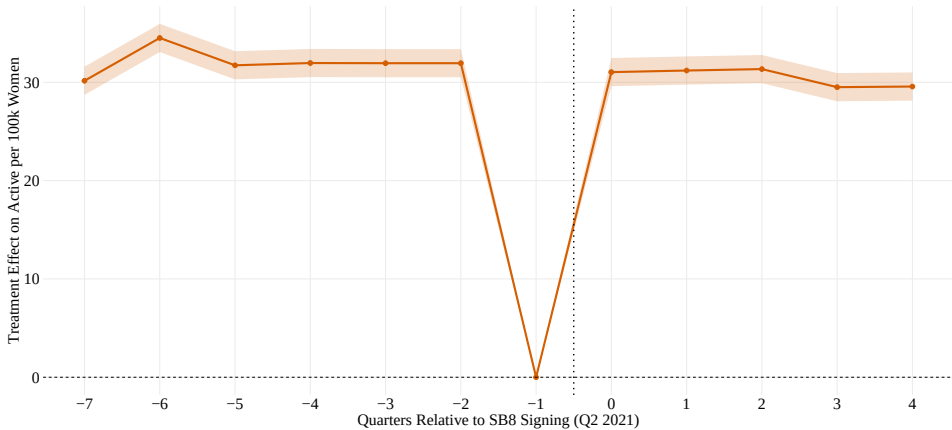
OLS. SE in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

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Full Results: ER Only

	<i>Dependent variable:</i>			
	ERs Active	(Per 100k Women) Exit State	(Per 100k Women) Enter State	Practice Exit
	(1)	(2)	(3)	(4)
in_TX	−15.28*** (2.29)	−0.24*** (0.04)	−0.26*** (0.07)	−0.01 (0.01)

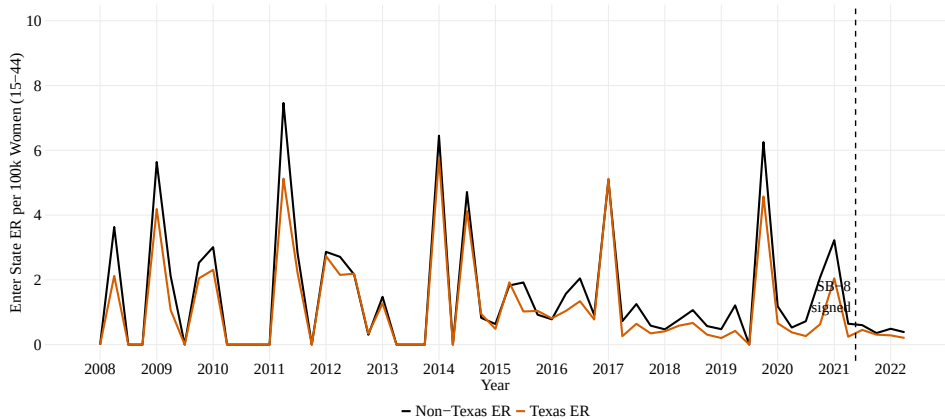
Event Study: Active ER



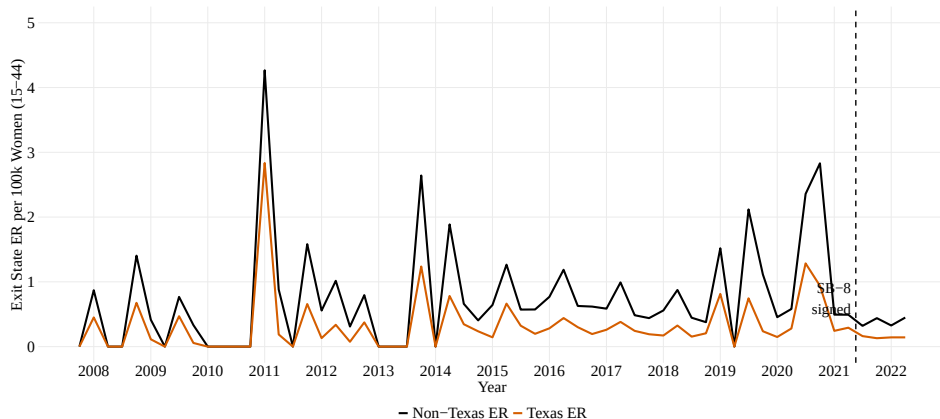
Synthetic Diff-in-Diff: ER Only

	<i>Dependent variable:</i>				
	ER (Per 100k Women)				
	Active	Exit State	Enter State	Exit Practice	Enter Pract
	(1)	(2)	(3)	(4)	(5)
ATT	0.70	−0.15	−0.25	0.00	0.001

Enter State: ER



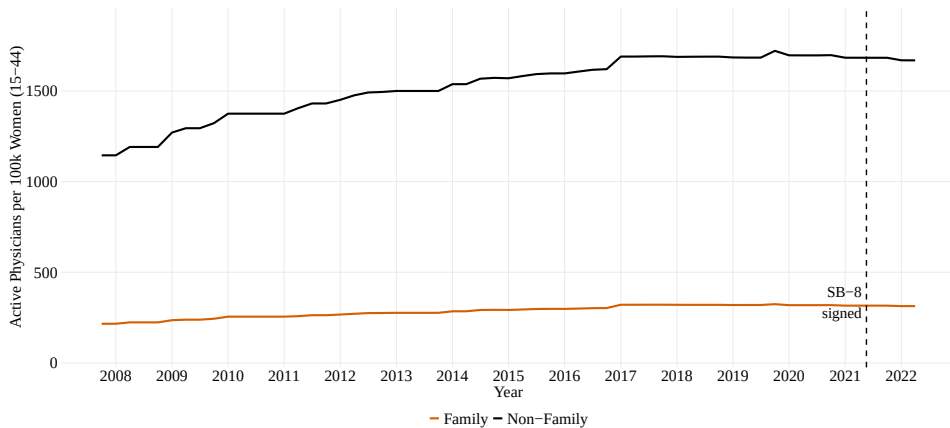
Exit State: ER



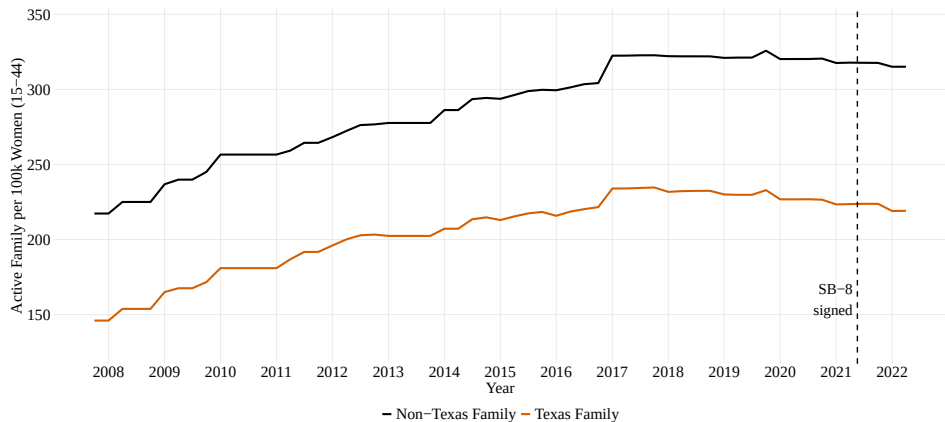
Summary Statistics: Family Medicine Only

	Mean	SD	Min	Max
Active OB/GYN	92.86	22.96	44	216
Active all other physicians	53.41	100.33	0	1248
Enter state OB/GYN	1.36	2.24	0	44
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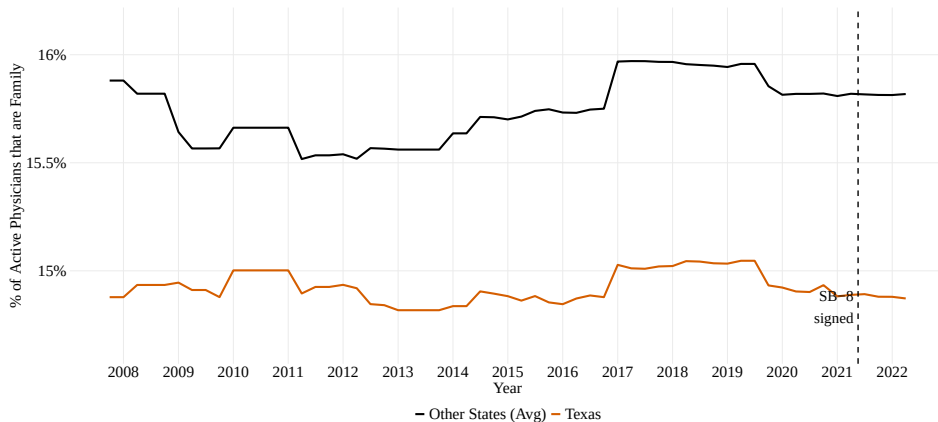
Trend: Family Medicine vs. Other Physicians



Trend: Family Medicine outside vs. inside Texas



Percentage of Active Physicians: Family Medicine



Descriptive Means: Family Medicine Only

	Not TX (Pre)	TX (Pre)	Not TX (Post)	TX (Post)
OB/GYN: Active	91.63	79.21	98.20	79.75
OB/GYN: Enter state	1.38	1.08	1.35	0.78
OB/GYN: Exit state	0.73	0.38	1.64	1.01
All other physicians: Active	52.95	37.32	56.35	38.71
All other physicians: Enter state	0.85	0.58	0.89	0.54
All other physicians: Exit state	0.45	0.20	1.09	0.71

Results: Family Medicine Only

	Active	Exit State	Enter State	Exit Practice	Enter Practice
β_1	-12.92 (42.87)	0.50 (0.75)	0.67 (1.32)	0.03 (0.20)	0.31 (0.97)
N	100,834	100,834	99,183	100,834	99,183

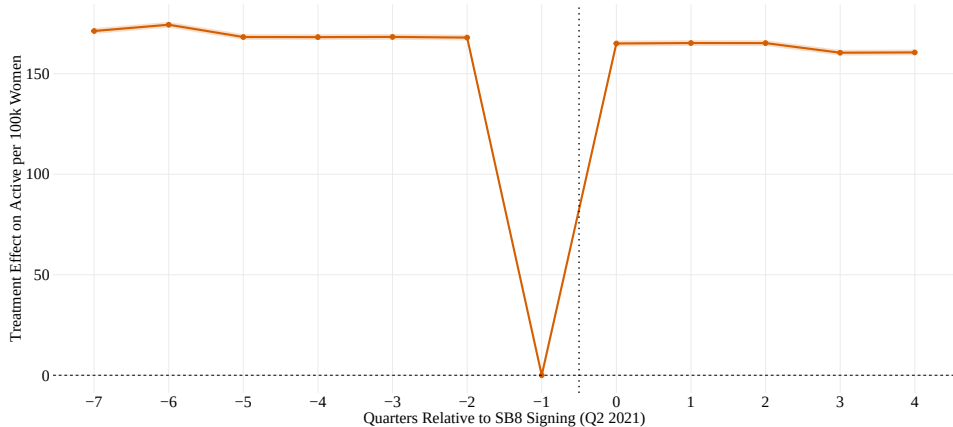
OLS. SE in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

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Full Results: Family Medicine Only

	<i>Dependent variable:</i>			
	Familys Active	(Per 100k Women) Exit State	(Per 100k Women) Enter State	Practice Exit
	(1)	(2)	(3)	(4)
in_TX	−13.29*** (2.10)	−0.22*** (0.04)	−0.23*** (0.07)	−0.01 (0.01)

Event Study: Active Family Medicine

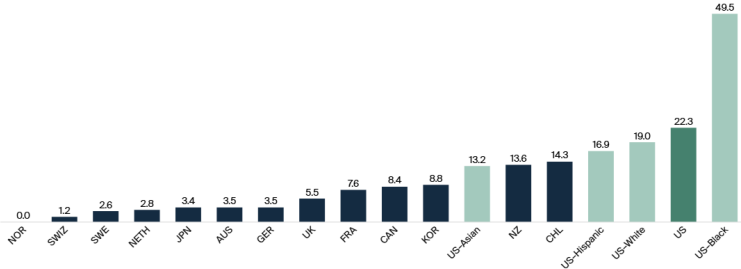


Synthetic Diff-in-Diff: Family Medicine Only

<i>Dependent variable:</i>					
Family (Per 100k Women)					
	Active	Exit State	Enter State	Exit Practice	Enter Practice
	(1)	(2)	(3)	(4)	(5)
ATT	6.01	−0.71	−0.87	0.001	0.002

The United States continues to have the highest maternal death rate, with the rate for Black women by far the highest of any group.

Maternal deaths per 100,000 live births



Notes: The maternal mortality ratio is defined as the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration and site of the pregnancy from any cause related to or aggravated by the pregnancy or its management but not from accidental or incidental causes. For more information on how maternal mortality is defined, see Organisation for Economic Co-operation and Development, "Maternal and Infant Mortality," in *Health at a Glance 2023: OECD Indicators* (OECD, 2023). 2015 data for FRA; 2017 data for UK; 2018 data for NZ; 2020 data for CAN and SWITZ; 2021 data for AUS, GER, JPN, KOR, NETH, and SWE; 2022 data for CHL (provisional), NOR, and US. Due to sample size limitations, data for US-AIAN cannot be displayed. AIAN = American Indian and Alaska Native. Asian Americans include a wide range of distinct communities. Such groupings are imperfect, as they mask significant difference in maternal mortality rates.

Data: All country data from OECD Health Statistics 2023 extracted on February 29, 2024, except data for US are 2022 data from the Centers for Disease Control and Prevention, National Center for Health Statistics, National Vital Statistics System, mortality and natality data files, "Maternal Mortality Rates in the United States, 2022."

Source: Munira Z. Gunja et al., *Insights into the U.S. Maternal Mortality Crisis: An International Comparison* (Commonwealth Fund, June 2024). <https://doi.org/10.26099/cthn-st75>

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